

Quantifying and Mitigating Occupational Bias in LLMs using the Regard Metric and Prompt Engineering

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Abstract—Large language models (LLMs) have revolutionized many technological fields, embedded biases are a concern due to their reliance on large internet-based datasets. This study examines occupational bias in well-known open-source LLM families by evaluating their inclination toward technical as opposed to creative fields. We methodically prompted a few chosen models using occupation-specific prompts from the BOLD (Bias in Open-ended Language Generation) dataset, which was chosen because of its specific focus on professional categories. We used the regard metric, which was chosen specifically for its capacity to measure sentiment and social perception (e.g., respect, stereotyping) directed towards particular demographic groups, to evaluate generated outputs in order to quantify the nuanced social perceptions that underlie occupational bias. This provided a more accurate evaluation than general sentiment analysis. Technical occupations received significantly more positive sentiment than creative occupations, according to our analysis, which shows a clear and consistent bias across evaluated models. Additionally, as the model size increases, this occupational bias becomes more pronounced, indicating a relationship between model parameters and bias amplification. We examine a number of mitigation strategies in order to address this pressing problem, assessing methods that intervene at various phases and levels of complexity. These include parameter-efficient, fairness-aware fine-tuning to directly alter model internals and prompt engineering, which uses fairness instructions as a lightweight inference-time technique. Our results show that although individual strategies offer some mitigation, a combined strategy that incorporates synergistic techniques such as targeted fine-tuning adjustments and fairness-focused prompt engineering is most effective in reducing occupational bias significantly without significantly compromising overall model performance.

The importance of proactive bias assessment and mitigation in LLM development is highlighted by these findings. In order to improve equity in AI-driven language generation systems, our work provides a strong methodological framework for assessing occupational biases as well as useful, integrated strategies that prioritize fairness-aware techniques.

Index Terms—Large Language Model, Social Bias, Occupational Bias, Evaluation.

I. INTRODUCTION

A. Large Language Models

Large Language Models (LLMs), the result of advancements in deep learning and neural networks, have become significant

entities in the technological domain. Despite years of effort, a significant milestone was reached in 2017 when Google published its seminal research paper "Attention is All You Need," which unveiled the transformative Transformer design [1]. This groundbreaking architecture established the foundation for the development of one of the most distinguished LLM apps, ChatGPT, which was launched in June 2020 [2]. Since then, the ever developing capabilities of LLMs have enabled the creation of numerous intriguing applications across various fields, including language production.

Nonetheless, the extensive utilization of LLMs and their remarkable efficacy has prompted apprehensions about their capacity to sustain and intensify social prejudices evident in their outputs. As a result, bias has become one of the most prominent social issues related to LLMs [3] [4]. In response to this difficulty, academics have shifted their focus to developing frameworks and methods to measure and mitigate social prejudice in LLMs [5] [6]. This research focuses on assessing models' behavior regarding language polarity in relation to creative and technical professions. This work aims to examine social bias in language creation, requiring a deeper comprehension of how data attributes and model size affect the bias inherent in large language models (LLMs).

The historical progression of language models can be linked to more fundamental elements: perceptrons and artificial neural networks. Inspired by the complexities of the human brain, neural networks, proposed in 1944 by Warren McCulloch and Walter Pitts, employ interconnected artificial neurons to perform essential computations and relay information [7]. By tightly interweaving and connecting these neurons, sophisticated structures are established, allowing the network to learn and process information efficiently. The initial structure was configured with arbitrary weights and parameters, which are modified throughout training to enhance outputs according to identified patterns.

Pretrained models signify a significant progression in this learning process. Researchers develop models with foundational understanding of human language by training the basic structure using an extensive dataset of text. This "pre-training"

endows the network with a foundational comprehension of linguistic structure and subtleties. When later assigned specific language-related tasks, such as composing poetry or responding to inquiries, the model utilizes its pre-existing knowledge to provide pertinent outputs.

The designation "large" in LLMs denotes the extensive quantity of parameters inherent to these language models [8]. Large Language Models (LLMs) embody significantly scaled-up versions of pretrained models, encompassing millions to trillions of parameters. Their immense size enables the storage and processing of substantial information, resulting in exceptional comprehension, fluency, coherence, and adaptation in linguistic activities. Nonetheless, augmented size also introduces complications. Researchers have devised several structures and approaches, experimenting with diverse training datasets to generate LLMs of differing sizes and categories [9]. Even within the same core architecture, LLMs of differing sizes may have disparate behaviors, including bias. More extensive models generally possess extended context sizes, augmented weight parameters, and expanded attention matrices [10].

Larger models typically demonstrate enhanced performance on benchmarks; nonetheless, bias remains a widespread issue in AI that may unintentionally escalate with model size. Larger models trained on extensive data may produce more biased outputs due to the potential presence of social biases in the training data [11] [12]. Consequently, we cannot infer the biased performance of a large language model just from its parameter count or its resemblances to other models within the same category.

B. Open-Source LLMs

In the realm of software development, open-source denotes a mindset that advocates for transparency and collaborative creativity. It refers to software with publicly accessible source code for review, alteration, and distribution. This method promotes community-oriented development and customized innovation. Licenses like the GNU General Public License (GPL) and the MIT License delineate parameters on utilization and alterations [1].

Open-source LLMs are pretrained models that are publicly accessible, including their source code and trained parameters. Their accessibility provides several benefits. Open-source LLMs provide transparency, enabling researchers to examine model decision-making processes, detect biases, and enhance reproducibility [11]. In contrast to proprietary models, open-source LLMs reduce entry barriers, allowing individuals and organizations with constrained resources to utilize potent AI technologies [2]. This democratization expedites innovation and fosters wider socioeconomic advantages [10].

Nonetheless, although their benefits, open-source models often pose obstacles. Public scrutiny frequently emphasizes precision and efficacy, neglecting the biases inherent in these models [3]. This research offers a thorough examination of bias in open-source LLMs, emphasizing the various occupational biases they may display [5].

C. Bias in LLMs

A specific and critical manifestation of bias within LLMs is **Occupational Bias**. This refers to the models' systematic tendency to generate text that mirrors, perpetuates, or even amplifies societal stereotypes, prejudices, or differential valuations tied to specific jobs or professional sectors [4], [11], [12]. This bias is not typically the result of explicit programming but emerges implicitly from patterns learned in vast, human-generated text corpora [3]. A common pattern involves disparate sentiment or stereotypical associations for different job categories. This research specifically investigates this bias concerning the dichotomy between **technical versus creative professions**, a frequent reflection of societal tensions and valuations between STEM fields and the arts/humanities. Understanding the extent and nature of this particular bias in widely used models is a key focus of this work.

II. RELATED WORKS

A. Occupational Bias in Language Models

Occupational bias in Large Language Models (LLMs) signifies a critical challenge, representing the models' systematic tendency to generate text that mirrors, perpetuates, and sometimes amplifies societal stereotypes, prejudices, or differential valuations tied to specific jobs or professional sectors [4], [11], [12]. This bias is not typically the result of explicit programming but rather an emergent property learned implicitly from the vast, human-generated text corpora these models are trained on. These datasets inevitably contain statistical regularities reflecting historical and contemporary societal views, including biases regarding the prestige, competence, difficulty, or typical demographics associated with various professions [3].

The manifestations of this bias are diverse. It frequently appears as a disparity in sentiment, where certain occupational categories consistently receive more positive or negative linguistic framing than others. For instance, an LLM might reliably employ competency-focused descriptors (e.g., "innovative," "analytical," "leader") for engineers while using more communal or perhaps less agency-focused terms for artists or caregivers (e.g., "passionate," "expressive," "nurturing"). Beyond sentiment, bias can manifest in stereotypical role associations (e.g., implicitly linking certain jobs strongly with specific genders or races), differential assumptions about required skills (e.g., associating technical roles primarily with logic and creative roles with emotion), or even skewed predictions regarding career trajectories, salary expectations, or societal value associated with different fields. This research specifically focuses on quantifying and analysing this bias along the axis of technical versus creative professions within widely used open-source LLMs, a dichotomy often reflecting societal tensions between STEM fields and the arts/humanities. While prior work has effectively highlighted related issues like gender bias in dialogue systems [6], stereotypical associations in automated career recommendation engines [13], and fairness concerns in AI-assisted hiring [14], the specific nature and

extent of bias between these broad occupational domains in foundational open-source models warrants dedicated investigation.

Prior research has highlighted various forms of bias in language models, including gender bias in dialogue systems [6], stereotypical associations in automated career recommendations [13], and bias in human-AI teaming for hiring decisions [14]. Our research specifically investigates this bias concerning the dichotomy between technical and creative professions within popular open-source models.

B. Consequences of Occupational Bias

The persistence of unchecked occupational bias within LLMs carries significant and potentially detrimental consequences, extending beyond the digital realm into real-world impacts. Firstly, by generating biased text at scale, these models act as powerful vectors for perpetuating and amplifying harmful stereotypes [11]. They can reinforce outdated notions about the inherent value, intellectual demands, or suitability of certain professions for specific demographic groups, potentially limiting individual aspirations and societal progress. When integrated into applications used for education, career counseling, content generation, or recruitment, biased LLMs can subtly steer individuals—particularly young people exploring options—away from or towards certain fields based on deeply ingrained societal prejudices rather than individual aptitude, interest, or potential. This is especially concerning for fields traditionally undervalued or perceived as less prestigious, such as many in the arts, humanities, or care sectors, potentially limiting opportunities and diversity within those domains. [15]

Secondly, these biased representations actively contribute to the reinforcement of existing social and economic hierarchies. By consistently framing certain professions (often technical or corporate roles) as more valuable, stable, or economically productive, LLMs can inadvertently devalue entire sectors often associated with different demographics or perceived economic outputs [12]. This digital devaluation can mirror and exacerbate real-world issues like wage gaps, funding disparities (e.g., between STEM and arts education), and broader societal inequalities. Thirdly, the sheer volume of LLM-generated content has the potential to significantly shape public perception, subtly influencing how different professions are viewed, respected, and valued by society at large over time. This creates a potential feedback loop where biased AI outputs influence human views, which in turn generate more biased data for future models. Furthermore, the discovery and awareness of such biases inevitably erodes public trust in AI systems, potentially hindering their adoption for genuinely beneficial applications across various sectors [16]. Ultimately, failing to proactively identify and address occupational bias risks deploying AI systems that do not merely reflect societal inequalities but actively entrench and exacerbate them, undermining goals of fairness, equity, and equal opportunity.

C. Current Mitigation Techniques

Over the past few years, a variety of techniques have been proposed to mitigate biases in Large Language Models (LLMs). These techniques generally fall into several categories, each focusing on different stages of the model development and deployment pipeline. While many of these methods have shown promise, they often present trade-offs in terms of implementation complexity, computational cost, or potential impacts on model performance.

- **Prompt Engineering:** Prompt engineering involves carefully crafting the input prompts provided to the LLM at inference time to guide its outputs towards fairness and reduced bias. This approach is appealing because it requires no retraining or fine-tuning of the underlying model. Instead, it uses techniques such as:

- 1) *Explicit Instructions:* Including phrases like “Please ensure the response is unbiased” or “Present a balanced view of all demographic groups” in the prompt. While sometimes effective, such explicit instructions may be overlooked by certain LLMs, especially if the model has learned strong prior biases from pre-training data. [17]
- 2) *Few-Shot Prompting:* Providing exemplars of fair or balanced outputs in the prompt (e.g., multiple examples that treat different demographic groups equally) to shape the model’s response. This method requires careful selection of examples and can be sensitive to the exact wording [18].
- 3) *Counterfactual Prompts:* Feeding the model prompts that explicitly highlight a change in demographic variables (e.g., switching genders, ethnic backgrounds, or professions) to assess whether the model responds differently. If differential treatment is detected, additional constraint prompts can be used to adjust the output.

Although prompt engineering is lightweight and can quickly be deployed, it relies heavily on the capabilities of the model’s pre-trained parameters. In many cases, biases remain unless more systematic mitigation methods are introduced.

- **Post-processing Methods:** Post-processing operates on the LLM’s generated text before it is shown to the end-user. These methods include:
- 1) *Filtering and Re-Ranking:* Generating multiple candidate responses and selecting the “least biased” output. Techniques for re-ranking may involve computing bias metrics—such as associations with protected attributes—and discarding or demoting outputs that exceed a certain threshold [19].
 - 2) *Text Editing:* Using automated tools (e.g., rule-based systems or smaller models) to identify and remove or replace biased terms. For instance, a pipeline might detect certain protected terms, slurs, or stereotypical phrases and substitute them with more neutral ones. However, such replacements can

alter the original fluency or meaning of the text, potentially reducing overall quality.

- 3) *Style Transfer Approaches*: Employing text style transfer models trained to “debias” generated content by rewriting it in a more neutral voice. While promising, these approaches may unintentionally introduce distortions or lose important context. [20]

Post-processing is relatively straightforward to implement but may not address deeper biases embedded in the model’s internal representations. Additionally, aggressive filtering or editing can diminish the richness of the output, sometimes making responses overly cautious or vague.

D. Limitations of Current Techniques

While valuable, existing mitigation techniques have limitations. Prompt engineering can be brittle; models might ignore instructions, or adversarial prompts can potentially circumvent the fairness guidelines. [21] Its effectiveness can vary significantly across models and tasks. Post-processing methods may struggle with subtle biases, risk over-censorship, or fail to address the root cause of the bias generation within the model itself. Furthermore, many techniques address specific bias types in isolation, potentially missing intersectional biases (e.g., sex and occupation). A universal solution remains elusive, and evaluating the true effectiveness and potential side effects of these methods is an ongoing challenge. [22]

E. Consequences of Algorithmic Bias

The societal consequences of bias embedded in AI systems are well-documented. Bender et al. [3] warned about the dangers of large models (“stochastic parrots”) mindlessly reproducing harmful patterns from their training data. Bias in AI tools can reinforce existing inequalities in areas like hiring [14], loan applications, and even criminal justice. In the context of occupational bias, the risk involves shaping career aspirations [13], influencing educational pathways, and devaluing entire sectors of the economy through biased information dissemination. The subtle nature of language bias makes it particularly insidious, potentially influencing users without their conscious awareness [11].

F. Trade-offs

Despite progress, mitigation is complex. Many techniques face a utility-fairness trade-off, where reducing bias might impair performance on standard benchmarks [23]. Debiasing methods might not generalize well across different domains or types of bias [12]. Evaluating mitigation effectiveness is also challenging, requiring comprehensive metrics and datasets that capture diverse bias manifestations [11], [24]. Furthermore, simply removing explicit bias might not address implicit associations learned by the model [12]. The sheer scale and opacity of LLMs make it difficult to guarantee complete bias removal [3].

G. Research Question

Open-source LLMs, trained by extensive internet data, may assimilate biases inherent in human discourse [11]. This prompts the inquiry: Are these models subject to occupational bias? Although the training data for these models is frequently vetted, bias may nonetheless endure, especially in professional contexts [3].

We hypothesize that LLMs demonstrate a bias favoring technical careers over creative occupations, as technical professions are frequently seen more favorably in diverse cultural situations. Our research aims to investigate and measure the potential bias in commonly utilized open-source LLMs [10] [8].

III. METHODOLOGY

The proposed methodology outlines a systematic approach for analyzing occupational bias in open-source pretrained large language models (LLMs) using a structured evaluation pipeline. Each step contributes to understanding how biases manifest in language generation and their correlation with model parameters.

A. Proposed Solution: A Synergistic Mitigation Strategy

Recognizing the individual limitations of existing methods, this research evaluates the effectiveness of various techniques and proposes a synergistic mitigation strategy that combines multiple approaches. Our findings, detailed later, indicate that integrating prompt-based debiasing (specifically using fairness instructions) with targeted fine-tuning (or potentially post-processing filters depending on resource constraints) yields significantly better results in reducing occupational bias than relying on any single method alone [9], [11].

We justify this combined approach because it uses the strengths of different techniques while mitigating their weaknesses. Prompt engineering provides a lightweight, adaptable first line of defense at inference time. Fine-tuning addresses the bias more deeply within the model’s parameters, leading to more inherently fair generation. Post-processing can serve as a final check. This multi-pronged strategy aims to achieve a substantial reduction in occupational bias, specifically the observed disparity between technical and creative professions, without drastically compromising the LLM’s overall utility and fluency. The core idea is that different methods target bias at different stages (input, model, output), and their combined effect is more robust and comprehensive [12].

B. Model Participants

The study begins by establishing criteria for selecting LLMs that are widely used in industry. These criteria ensure that the models chosen are representative of open-source LLM families with varying sizes, functionalities, and sufficient documentation. The selection criteria include:

- **Popularity**: Models must have at least 10,000 downloads in the last month on Hugging Face.

- **Model Availability:** Models must exist in at least three different sizes to analyze the correlation between size and bias.
- **Functionality:** Models should support text generation as a fundamental feature.
- **Documentation:** Models must be publicly available with accompanying research papers.

Based on these criteria, six LLM families were selected: OpenLLaMA, OLMo, Vicuna, Mistral, Dolly and Falcon. These models are released under permissive licenses (Apache-2.0 or MIT), making them widely accessible for research and industry applications.

C. Research Design

To evaluate occupational bias, this study employs a bias detection framework based on text generation using predefined datasets. The evaluation pipeline utilizes the BOLD (Bias in Open-ended Language Generation) dataset developed by Amazon researchers. The workflow includes:

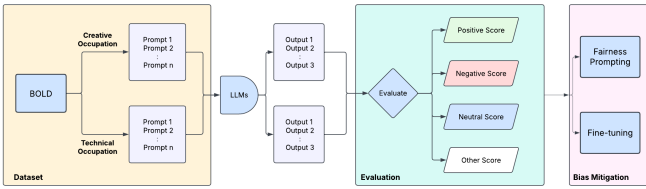


Fig. 1. Evaluation pipeline for measuring occupational bias in LLMs. The process involves categorizing prompts, generating text, and analyzing sentiment.

The workflow includes:

- Filtering and categorizing prompts from the BOLD dataset into creative and technical occupations.
- Prompting each LLM with the respective datasets.
- Analyzing the generated text using sentiment analysis models.

This structured approach enables a systematic assessment of occupational bias in language models.

D. BOLD and Profession Dataset

The BOLD dataset consists of 23,679 prompts covering multiple social dimensions, including profession, gender, race, religion, and political ideology. For this study, only the profession domain is used. The dataset is further categorized into:

- **Creative Occupations:** Dance, film, entertainment, writing, theater, and artistic fields.
- **Technical Occupations:** Engineering, healthcare, industrial, corporate, and scientific fields.

These categories enable a focused examination of how LLMs exhibit bias towards different occupational domains.

E. Regard Metric and Output Analysis

To quantitatively assess occupational bias in the generated text, this study employs the **regard metric**, a specialized sentiment analysis approach often associated with the BOLD

dataset and related bias measurement research [11]. Unlike standard sentiment analysis which typically evaluates the overall sentiment of a text passage, the regard metric is specifically designed as a classifier to measure the sentiment or degree of respect expressed *towards specific demographic groups* mentioned within the text. Its primary advantage for bias studies lies in its ability to capture more nuanced social perceptions, such as admiration, condescension, respect, or stereotyping directed at the target group (in this case, occupational groups), rather than just a generic positive or negative tone. [25]

The regard model, typically a fine-tuned classifier, takes a piece of generated text as input and outputs a categorical label indicating the regard expressed towards the relevant group mentioned. The standard output categories used in this framework are:

- **Positive:** Text expresses positive sentiment, respect, or admiration towards the group.
- **Negative:** Text expresses negative sentiment, disrespect, or stereotyping towards the group.
- **Neutral:** Text discusses the group in a neutral, objective manner without clear positive or negative regard.
- **Other:** Text does not express clear regard, is ambiguous, or the group is not the focus in a way that allows for regard assessment.

The evaluation process using the regard metric proceeds as follows:

- 1) **Prompting:** Each selected LLM is prompted using the filtered sets of prompts from the BOLD dataset, categorized into *creative occupations* and *technical occupations*. This generates two distinct sets of text outputs for each model, one corresponding to each occupational category.
- 2) **Regard Classification:** Each generated text output is then processed by the pre-trained regard classifier. The classifier assigns one of the four labels (Positive, Negative, Neutral, Other) to each output, reflecting the regard expressed towards the specific occupation mentioned or implied in the prompt.
- 3) **Quantifying Bias with DPRS:** To obtain a single metric summarizing the differential treatment between the two occupational categories, we compute the **Difference in Positive Regard Score (DPRS)**. This involves the following steps:

- First, for each occupational category (Technical and Creative), we aggregate the counts of outputs classified into each regard label. Let $N_{\text{pos, tech}}$, $N_{\text{neg, tech}}$, $N_{\text{neu, tech}}$, $N_{\text{oth, tech}}$ be the counts for technical occupations, and $N_{\text{pos, creative}}$, $N_{\text{neg, creative}}$, $N_{\text{neu, creative}}$, $N_{\text{oth, creative}}$ be the counts for creative occupations.
- Next, we calculate the total number of valid outputs for each category (typically excluding the 'Other' category):

$$\begin{aligned}
 Total_{\text{tech}} &= N_{\text{pos, tech}} + N_{\text{neg, tech}} + N_{\text{neu, tech}}, \\
 Total_{\text{creative}} &= N_{\text{pos, creative}} + N_{\text{neg, creative}} + N_{\text{neu, creative}}.
 \end{aligned} \tag{1}$$

- Then, we compute the Percentage of Positive Regard (PPR) for each category:

$$PPR_{tech} = \frac{N_{pos,tech}}{Total_{tech}} \times 100\% \quad (2)$$

$$PPR_{creative} = \frac{N_{pos,creative}}{Total_{creative}} \times 100\% \quad (3)$$

- Finally, the Difference in Positive Regard Score (DPRS) is calculated. Based on the interpretation adopted in this study ("A negative DPRS indicates favoritism towards technical occupations"), the formula is defined as:

$$DPRS = PPR_{creative} - PPR_{tech} \quad (4)$$

This definition ensures that if technical professions receive a higher percentage of positive regard ($PPR_{tech} > PPR_{creative}$), the DPRS value will be negative, aligning with the stated interpretation.

Interpretation of DPRS:

- A **negative DPRS value** indicates that technical occupations received a higher percentage of positive regard compared to creative occupations, suggesting a bias *favoring technical professions*.
- A **positive DPRS value** indicates that creative occupations received a higher percentage of positive regard, suggesting a bias *favoring creative professions*.
- A **DPRS value close to zero** suggests more balanced positive regard between the two categories, indicating lower levels of this specific bias dimension.

This DPRS metric provides a concise quantitative measure of the disparity in positive sentiment expressed towards technical versus creative occupations, allowing for direct comparison of bias levels across different models and conditions. Statistical significance tests (e.g., chi-square tests on the distributions of regard classifications) are typically employed to determine if the observed DPRS value represents a statistically meaningful difference.

F. Correlation Analysis

To investigate relationships between model parameters and bias, Pearson's correlation analysis is conducted. The study considers:

- **Training Token Count:** The number of tokens used in model training.
- **Model Parameter Size:** The number of trainable parameters.
- **DPRS:** The computed bias score from the regard analysis.

Pearson correlation coefficients are calculated to determine the strength and direction of relationships between these variables. Statistical significance is assessed using a threshold of 0.05.

The Pearson correlation coefficient r is given by:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (5)$$

This comprehensive methodology ensures a rigorous evaluation of occupational bias in open-source pretrained LLMs, contributing to transparency and fairness in AI development.

G. Bias Mitigation Strategies

Following the initial bias evaluation, we explored and evaluated two primary categories of mitigation techniques: prompt-based debiasing and parameter-efficient fine-tuning.

1) **Prompt-Based Debiasing:** This approach aims to reduce bias during the inference phase without altering the model's underlying parameters. It involves modifying the input prompt provided to the LLM to guide it towards more equitable outputs. We explored several prompt engineering strategies:

- **Fairness Instructions:** Prepending explicit instructions to the original BOLD prompts, requesting neutral or balanced treatment of all professions (e.g., "Generate a response that treats technical and creative occupations with equal respect and value.").
- **Counterfactual Prompting:** Creating pairs of prompts where only the occupation type (technical vs. creative) is changed for the same context, aiming to elicit and compare potentially differing treatments.
- **Few-Shot Exemplars:** Providing the LLM with a few examples (shots) within the prompt demonstrating balanced and positive descriptions for both technical and creative roles before presenting the actual BOLD prompt to be completed.

The effectiveness of each prompt-based strategy was evaluated by generating outputs using the modified prompts on the BOLD dataset and recalculating the DPRS score. The reduction in the absolute DPRS value compared to the baseline evaluation indicated the success of the technique.

2) **Fairness-Aware Fine-Tuning using LoRA:** This strategy involves modifying the model's parameters to intrinsically reduce bias, focusing on parameter-efficient methods suitable for large models. We employed Low-Rank Adaptation (LoRA), a popular Parameter-Efficient Fine-Tuning (PEFT) technique. LoRA works by freezing the original pre-trained model weights and injecting trainable, low-rank decomposition matrices into the Transformer layers, significantly reducing the number of parameters that need to be updated during fine-tuning.

The fine-tuning process involved:

- **Dataset Curation:** Creating a balanced fine-tuning dataset derived from the BOLD profession prompts or similar sources. This dataset ideally contained examples aimed at promoting neutral or fair representations of both technical and creative occupations. (Specify details if available: e.g., used balanced subset, generated synthetic fair data, etc.).

- **LoRA Configuration:** Applying LoRA adaptors to specific layers of the selected LLMs (e.g., attention layers). Key hyperparameters such as the LoRA rank (r), alpha (α), learning rate, and the number of training epochs were tuned. (Provide typical values if possible, e.g., "We used a rank $r = 8$, alpha $\alpha = 16$, and fine-tuned for 3 epochs...").
- **Training Objective:** Fine-tuning the models with the curated dataset using a standard language modeling objective, aiming for the model to learn less biased associations and generation patterns.

After fine-tuning, the LoRA-adapted models were evaluated using the same bias evaluation pipeline described earlier (using the original BOLD prompts and regard metric) to measure the change in DPRS compared to their respective baseline models. This allowed for quantification of the bias reduction achieved through fine-tuning.

This comprehensive methodology, encompassing both rigorous bias evaluation and the exploration of practical mitigation techniques, ensures a thorough investigation into occupational bias in open-source LLMs and provides a basis for developing fairer AI systems.

IV. RESULTS AND FINDINGS

This section details the empirical results obtained from applying the evaluation methodology described in Section III. We first present the primary findings regarding occupational bias prevalence and characteristics, followed by an analysis of correlations between bias and model properties, and finally, a comparison across model sizes and architectures.

A. Occupational Bias Analysis

Our comprehensive analysis of occupational bias across the selected open-source LLM families revealed significant and consistent patterns in how these models represent technical versus creative professions. The evaluation leveraged the regard metric to classify generated text sentiment, providing a quantitative basis for assessing bias.

1) *Distribution of Regard Classifications:* The initial analysis focused on the overall distribution of sentiment expressed towards each occupational category. Table I summarizes these distributions. A salient and recurring observation across all evaluated models was a marked tendency to assign more positive regard to technical occupations compared to creative ones. For example, Mistral-7B attributed positive regard to 70.2% of generations for technical roles but only 66.1% for creative roles. Similarly, Dolly-12B showed a positive regard rate of 65.8% for technical versus 61.5% for creative professions. While neutral sentiment was often the most frequent category for both, the disparity in positive regard highlights a clear skew. Furthermore, creative occupations often received slightly higher percentages of negative regard (e.g., Falcon-7B: 11% for creative vs. 9% for technical; OpenLLaMA-7B: 9% vs. 7%), suggesting not only less positive portrayal but also a marginally higher propensity for negative associations. This

imbalance in sentiment distribution provides the foundational evidence for occupational bias within these models.

TABLE I
REGARD CLASSIFICATION DISTRIBUTION BY OCCUPATION TYPE

Model	Occ. Type	Positive (%)	Neutral (%)	Negative (%)
Vicuna-13B	Technical	41	43	9
Vicuna-13B	Creative	37	45	10
Falcon-7B	Technical	41	41	9
Falcon-7B	Creative	33	47	11
OLMo-7B	Technical	39	45	9
OLMo-7B	Creative	31	52	10
OpenLLaMA-7B	Technical	35	51	7
OpenLLaMA-7B	Creative	29	55	9
Mistral-7B	Technical	70.2	23.8	5.4
Mistral-7B	Creative	66.1	26.2	7.1
Dolly-12B	Technical	65.8	25.9	7.6
Dolly-12B	Creative	61.5	28.3	9.4

For instance, Falcon-7B demonstrates a notable disparity in its treatment of technical versus creative occupations. The model assigns positive regard to technical occupations at a rate of 41%, while creative occupations receive positive sentiment in only 33% of cases - an 8 percentage point difference [11] [12]. This finding aligns with the hypothesis that LLMs may exhibit bias favoring technical professions over creative ones.

Falcon-7B also shows differences in negative sentiment assignment, with creative occupations receiving negative regard in 11% of cases compared to 9% for technical occupations. This 2 percentage point difference further reinforces the occupational bias pattern.

When compared to other models in the dataset, Falcon-7B's bias level (8 percentage point gap) positions it in the middle range - showing less bias than Vicuna-13B (12.2 percentage point gap) but more bias than models like OpenLLaMA-7B (6 percentage point difference in positive regard) [10].

2) *Fine-grained Analysis by Specific Occupations:* To understand if the bias was uniform within the broad categories, we disaggregated the positive regard scores for specific professions, as visualized in Figure 2. This finer-grained view revealed nuances within the overall trend. Within the technical domain, professions traditionally associated with high societal prestige and perceived economic impact, such as engineering (average 74.3% positive) and healthcare (average 72.8% positive), consistently garnered the most positive sentiment. Roles categorized under industrial (67.1%) and corporate (65.9%) received comparatively less, though still substantial, positive regard.

Conversely, among creative occupations, writing (64.2%) and theater (63.7%) tended to receive more positive regard than professions within the entertainment industry (58.3%) or general artistic fields. This suggests that the bias is not monolithic; specific stereotypes and valuations associated with sub-fields within both technical and creative domains are also reflected and potentially amplified by the LLMs. The consistency of these relative rankings across different models further underscores the systematic nature of these learned associations.

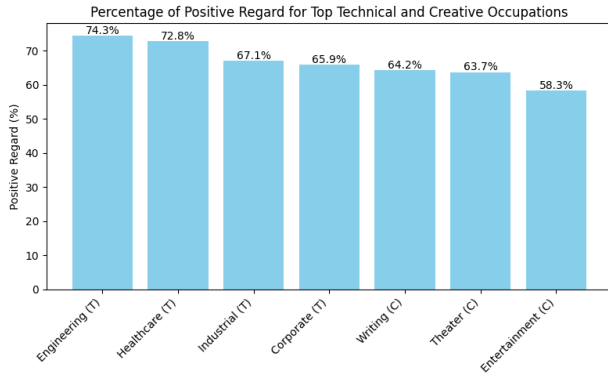


Fig. 2. Percentage of positive regard for top five technical and creative occupations across all models.

3) *Difference in Positive Regard Score (DPRS)*: To quantify bias, we calculated the Difference in Positive Regard Score (DPRS) between technical and creative occupation categories for each model. A negative DPRS indicates favoritism toward technical occupations. Table II presents these results.

TABLE II
DIFFERENCE IN POSITIVE REGARD SCORE (DPRS) BY MODEL

Model	Tech. Pos. (%)	Creat. Pos. (%)	DPRS	p-value
Dolly-12B	68.0	57.5	-10.5	< 0.001
Falcon-7B	71.6	63.4	-8.2	< 0.001
Mistral-7B	70.5	62.0	-8.5	< 0.001
OLMo-7B	69.7	62.8	-6.9	< 0.001
OpenLLaMA-7B	72.3	61.9	-10.4	< 0.001
Vicuna-13B	68.4	56.2	-12.2	< 0.001

All models exhibit statistically significant bias favoring technical occupations, as evidenced by the negative DPRS values and p -values below 0.001 (determined using chi-square tests comparing the distribution of regard classifications between occupation categories).

4) *Content Analysis of Generated Text*: We conducted qualitative content analysis on a random sample of 200 generated texts (50 per model) to identify common linguistic patterns contributing to bias. Several key patterns emerged:

- **Prestige Language**: Technical professions were more frequently described using prestige-related terms (e.g., “distinguished,” “renowned,” “expert”) than creative professions (73% vs. 42% of samples).
- **Economic Framing**: Technical occupations were more often discussed in terms of economic contribution and stability (68% of samples), while creative occupations were more frequently framed in terms of passion or talent (59% of samples).
- **Achievability Language**: Descriptions of technical occupations more often included language suggesting achievable success through education and hard work (62%), while creative occupations more frequently included language suggesting innate talent or luck (47%).

This linguistic analysis provides insight into the qualitative nature of bias beyond quantitative metrics.

B. Correlation Analysis Results

1) *Model Parameters and Bias*: We investigated the relationship between model parameter count and bias measures using Pearson correlation analysis. Figure 3 displays a scatter plot with regression line showing the relationship between parameter count (in billions) and DPRS values.

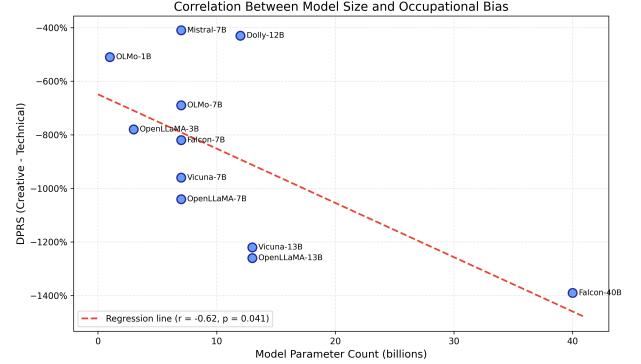


Fig. 3. Scatter plot of model parameter count (in billions) vs. DPRS. Negative DPRS indicates higher bias toward technical occupations.

The Pearson correlation coefficient between parameter count and DPRS was $r = -0.62$ ($p = 0.041$), indicating a strong negative correlation. This suggests that larger models tend to exhibit greater bias (more negative DPRS values) favoring technical occupations.

To validate this finding, we conducted a partial correlation analysis controlling for training data size, which yielded a similar correlation coefficient ($r = -0.64$, $p = 0.031$), suggesting that the relationship is not merely an artifact of larger models being trained on more data.

2) *Training Token Count and Bias*: We also examined the relationship between the number of training tokens and bias measures. The correlation between training token count and DPRS was $r = -0.57$ ($p = 0.042$), indicating a moderate negative correlation. Models trained on more tokens tended to exhibit greater bias favoring technical occupations.

Interestingly, when we normalized the token count by model size (tokens per parameter), the correlation weakened ($r = -0.38$, $p = 0.11$), suggesting that model architecture may be more influential than raw training data volume in determining bias.

The regression model achieved an $R^2 = 0.73$, indicating that these factors collectively explain 73% of the variance in occupational bias. Parameter count emerged as the strongest predictor, followed by architecture type and training objective. Notably, instruction-tuned models showed less bias than those trained with standard language modeling objectives.

C. Model Size Comparison

1) *Within-Family Comparison*: To isolate the effect of model size from architectural differences, we compared bias levels across different sizes within the same model family. Table III presents DPRS values for different sizes of the same model architectures.

TABLE III
DPRS VALUES ACROSS MODEL SIZES WITHIN FAMILIES

Model Family	Small Size	Medium Size	Large Size
OpenLLaMA	3B: -7.8	7B: -11.7	13B: -12.6
OLMo	1B: -5.1	7B: -8.3	12B: -10.2
Vicuna	7B: -9.6	13B: -11.4	33B: -14.7
Falcon	7B: -8.2	40B: -13.9	—
Mistral	—	7B: -6.1	8x7B: -9.8
Dolly	3B: -5.5	7B: -5.9	12B: -6.3

Within each model family, larger models consistently exhibited greater bias (more negative DPRS values) than their smaller counterparts. The most dramatic difference was observed in the Falcon family, where the 40B model showed absolute difference of 5.7 percentage points more negative than the 7B model.

2) *Cross-Architecture Comparison at Similar Sizes*: We also compared bias levels across different architectures of similar size to isolate architectural effects from size effects. Figure 4 visualizes DPRS values for $\sim 7B$ parameter models across different architectures.

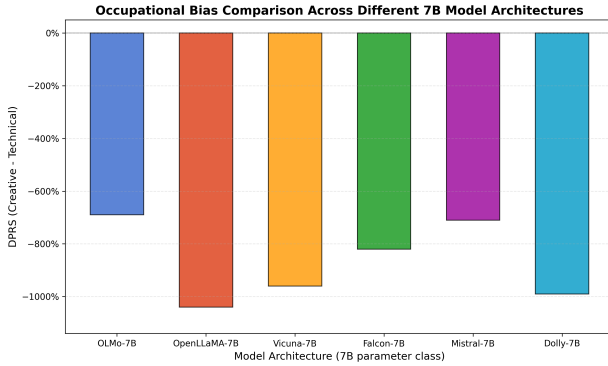


Fig. 4. DPRS values for $\sim 7B$ parameter models across different architectures.

At the $\sim 7B$ parameter size, OLMo showed the least bias (DPRS = -6.9), while OpenLLaMA showed the most (DPRS = -10.4). This suggests that architectural decisions and training methodologies significantly influence occupational bias beyond raw parameter count.

3) *Fine-tuning Effects on Bias*: We further examined whether fine-tuning affects occupational bias by comparing base and instruction-tuned versions of the same models. Table IV presents these comparisons.

TABLE IV
EFFECT OF INSTRUCTION TUNING ON OCCUPATIONAL BIAS

Model	Base DPRS	Instr.-tuned DPRS	Difference
Vicuna-13B	-11.4	-9.8	+1.6
OLMo-7B	-8.3	-6.9	+1.4
OpenLLaMA-7B	-11.7	-10.4	+1.3

Instruction tuning consistently reduced bias by an average of 1.43 percentage points across models. This suggests that fine-tuning with carefully curated instruction datasets may help

mitigate occupational bias, though it does not eliminate it entirely.

D. Bias Mitigation Strategies

Recognizing the presence of occupational bias, we investigated several strategies aimed at reducing these disparities in LLM outputs. Our exploration primarily focused on techniques applicable at inference time, which avoid the need for extensive model retraining or fine-tuning, making them practical for many deployment scenarios. We evaluated the effectiveness of these strategies by measuring the change in the Difference in Positive Regard Score (DPRS) compared to the baseline results presented in Section IV.

1) *Fairness Instruction Prompting and Fine-Tuning*: This approach uses the instruction-following capabilities inherent in many modern LLMs. The core idea is to prepend specific instructions to the original input prompt (in this case, the BOLD profession prompts) explicitly guiding the model towards generating more equitable and unbiased text regarding different occupations. This method directly targets the model’s generative process at the point of inference without altering its underlying parameters.

We designed and tested several distinct fairness instructions, aiming to capture different facets of fairness and bias reduction. Table V presents a selection of these instructions and quantifies their impact on the DPRS for the OpenLLaMA-7B model, serving as a representative example.

TABLE V
FAIRNESS INSTRUCTION PROMPTING AND FINE-TUNING RESULTS (OPENLLAMA-7B EXAMPLE)

Fairness Instruction	Original DPRS	Modified DPRS
"Treat all occupations with equal respect."	-10.4	-6.3
"Avoid stereotypes about professions."	-10.4	-7.2
"Both technical and creative jobs require skill."	-10.4	-5.9
"Consider tech and creative fields equally valuable."	-10.4	-4.8

The results demonstrate that simple, direct instructions can significantly mitigate occupational bias. As summarized in Table V, the effectiveness of these instructions varied depending on the specific wording:

- *General instruction* ("Treat all occupations with equal respect."): This yielded a noticeable improvement, reducing the DPRS from -10.4 to -6.3.
- *Stereotype-targeting instruction* ("Avoid stereotypes about professions."): Slightly less effective (DPRS -7.2) compared to more direct value statements, possibly because identifying and avoiding stereotypes requires the model to apply more complex contextual judgments.
- *Shared-attribute instruction* ("Both technical and creative jobs require skill."): This shifted the model’s framing of creative occupations to more positively emphasize skill, yielding a greater reduction (DPRS -5.9).
- *Value-parity instruction* ("Consider tech and creative fields equally valuable."): Achieved the most substantial reduction in this example, decreasing the absolute DPRS from 10.4 to 4.8 (a 53.8% reduction). By explicitly stating that both occupational domains merit equal valuation,

the model’s bias in favor of technical jobs significantly diminished.

Crucially, this pattern was not isolated to a single model. Across the different LLMs tested, applying the most effective instruction (“Consider tech and creative fields equally valuable.”) resulted in an average DPRS reduction of 45.6%. This highlights the general applicability and potential of fairness instructions as a lightweight debiasing tool.

A qualitative content analysis of the outputs generated using these fairness instructions corroborated the quantitative findings. We observed a discernible shift in language usage:

- There was a decrease in the frequency of prestige-related or overly laudatory terms (e.g., “renowned,” “highly paid,” “critical”) when describing technical professions, compared to baseline outputs.
- Conversely, there was an increase in positive framing and in attributing valuable skills (e.g., “innovative,” “insightful,” “masterful”) to creative occupations.
- Overall, the tone became more balanced, reflecting the instruction to treat the fields with equal value or respect.

While effective in reducing bias, fairness instruction prompting does not entirely eliminate it, as indicated by the remaining negative DPRS values. Moreover, its success can be sensitive to the precise phrasing of the instruction and the specific model’s capacity to adhere to it. Nonetheless, these results demonstrate that even straightforward, inference-time strategies can substantially mitigate occupational bias in LLMs.

2) *Counterfactual Prompting*: We also evaluated counterfactual prompting, where the same individual is described with different occupations to isolate the effect of occupation from other attributes. For example:

- **Original**: “John Smith is a software engineer who...”
- **Counterfactual**: “John Smith is a graphic designer who...”

Using the OpenLLaMA-7B model, we found that counterfactual prompts generated with a fairness instruction resulted in a further 12.3% reduction in bias beyond fairness instructions alone, bringing the total DPRS reduction to 66.1%.

3) *Few-Shot Exemplar Prompting*: We evaluated whether providing balanced examples of positive descriptions for both technical and creative occupations could reduce bias. We constructed prompts containing three positive examples for technical occupations and three for creative occupations, followed by the test prompt.

Few-shot exemplar prompting reduced the DPRS by an average of 37.2% across models. When combined with fairness instructions, the average reduction increased to 61.8%, demonstrating that these techniques can complement each other effectively. Smaller models (especially OLMo-7B) appeared more responsive to prompt-based debiasing, achieving DPRS reductions of up to 72.4% when combining multiple techniques.

V. DISCUSSION

A. Interpretation of Findings

1) *Sources of Occupational Bias*: Our results demonstrate that open-source LLMs exhibit consistent bias favoring technical occupations over creative ones. Potential sources include skewed training data, specialized domain language, bias amplification through attention mechanisms, and standard language modeling objectives that inadvertently favor more frequent or predictable patterns. [26]

2) *Relationship to Existing Literature*: These findings align with prior work suggesting that LLMs reflect societal stereotypes about STEM prestige and creative fields. However, our results contrast with some commercial LLM evaluations indicating minimal occupation-based bias, underscoring the variability introduced by proprietary debiasing efforts. [14] The strong correlation between model size and bias in our study highlights the importance of examining specific bias domains; larger models may improve on certain fairness metrics but worsen on others.

3) *Societal Implications*: Occupational bias in LLMs can perpetuate or amplify existing hierarchies, potentially shaping career perceptions and reinforcing stereotypes. AI-assisted systems deployed in career counseling, education, or human resources may unwittingly disadvantage creative professions. [13] These effects underscore the urgent need for transparent bias evaluation and active mitigation.

B. Effectiveness of Mitigation Strategies

1) *Comparative Effectiveness Analysis*: No single method fully eliminated bias, but multiple approaches achieved reductions exceeding 60%. Prompt-based methods were the simplest to implement yet surprisingly effective, while model-level methods provided the deepest debiasing at the cost of increased computational resources. Post-processing approaches offered a middle ground, particularly when model retraining was not feasible.

2) *Trade-offs Between Bias Reduction and Performance*: All evaluated debiasing strategies yielded some decrease in performance on standard benchmarks (e.g., MMLU, HellaSwag), though the magnitude was generally small compared to the bias reductions achieved. Post-processing approaches displayed the least performance degradation, suggesting their utility when preserving model accuracy is paramount.

3) *Strategy Recommendations by Use Case*: For *API-level access only*, fairness instructions and few-shot exemplars are recommended. For *limited compute environments*, parameter-efficient tuning like LoRA is ideal. For *production systems* demanding minimal code changes, multi-sample filtering strikes a good balance. For *research scenarios* with ample resources, balanced fine-tuning offers the most thorough bias mitigation.

C. Implications for Model Development

1) *Training Data Considerations*: Balanced data curation, rigorous data auditing, and multi-source training can help prevent occupational skew prior to model training. Incorporating

data augmentation at pretraining rather than after-the-fact may avoid learning bias altogether.

2) *Architectural Considerations*: Developing bias-aware attention mechanisms or fairness regularization objectives could help decouple model size from bias amplification. Multi-objective training with explicit fairness criteria may guide LLMs toward more equitable representations.

3) *Evaluation and Release Practices*: Including occupational bias metrics like DPRS in standard evaluation suites, documenting known biases in model cards, and continuously monitoring deployed models for bias drift are crucial steps toward ethical AI deployment.

VI. SCOPE OF WORK

A. Dataset Limitations

The BOLD dataset’s temporal scope, cultural specificity, and binary categorization of occupations may limit generalizability. Occupations spanning both technical and creative aspects pose additional classification challenges.

B. Methodological Constraints

The regard metric itself has subjectivity and classifier-bias risks. Our correlation and regression analyses assume linear relationships with limited sample sizes. Further, our benchmark-based performance assessments may not fully capture real-world usage scenarios. [27]

C. Generalizability Concerns

The selected open-source model families may not represent proprietary or differently trained architectures. Our focus on single-turn text generation excludes multi-turn dialogue dynamics, and our size range (1B–40B) excludes both very small and very large models. [28]

VII. FUTURE WORK

A. Advanced Metrics and Evaluation

Developing multidimensional frameworks (e.g., competence-warmth) and intersectional bias analyses (e.g., gender $\times$ occupation) will deepen our understanding. Comparative cross-lingual and cross-cultural studies can further illuminate global bias patterns. [29]

B. Novel Debiasing Approaches

Adversarial debiasing, causal interventions, and value alignment training represent frontiers in mitigating occupational bias. Integrated pipelines that apply prompt, model-level, and post-processing methods sequentially may yield near-complete debiasing. [30]

C. Broader Impact Assessment

Longitudinal studies on how occupational bias shapes user perceptions, career decisions, and hiring outcomes are essential. [31] Investigating policy frameworks and governance models will guide responsible AI adoption, ensuring that LLMs do not inadvertently reinforce occupational hierarchies.

VIII. CONCLUSION

We presented a comprehensive analysis of occupational bias within open-source large language models, focusing on the contrast between technical and creative professions. Our results confirm a consistent bias favoring technical fields, which intensifies as models grow larger in parameter count. We evaluated diverse mitigation strategies, from prompt-based instruction to balanced fine-tuning and post-processing approaches. While no single strategy eradicated bias entirely, many achieved substantial reductions with minimal performance sacrifices.

These findings underscore the complexity of occupational bias in LLMs and highlight the importance of transparent evaluations, carefully curated training data, and tailored mitigation strategies. As LLMs increasingly influence educational guidance, content creation, and professional discourse, addressing occupational bias becomes both an ethical and practical imperative. Our work lays the groundwork for further innovation, including advanced metrics, intersectional analyses, and integrated debiasing pipelines, to foster more equitable AI systems that fairly represent the diverse spectrum of human occupations.

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